

ZENITH ORBITAL SYSTEMS

IT Department Business Continuity Plan

Ground-to-Space Link Redundancy Edition

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Section I: Introduction

A. How to Use This Plan

In the event of a disaster which interferes with Zenith Orbital Systems' ability to conduct business from its primary Dallas ground station facility, this plan is to be used by the responsible individuals to coordinate the business recovery of their respective areas and departments. The plan is designed to contain, or provide reference to, all of the information that might be needed at the time of a business recovery.

This plan is not intended to cover the operations of the ZOS separately structured Emergency Response Team. Index of Acronyms: (EOC) Emergency Operations Center; (EMT) Emergency Management Team; (ERT) Emergency Response Team; (BCP) Business Continuity Plan; (IT) Information Technology; (ZOS) Zenith Orbital Systems; (GSL) Ground-to-Space Link.

Section I, Introduction, contains general statements about the organization of the plan. Section II, Business Continuity Strategy, describes the strategy the IT Department will implement to maintain business continuity in the event of a facility disruption. Section III, Recovery Teams, lists the Recovery Team functions, those individuals assigned specific responsibilities, and procedures on how each team member is to be notified. Section IV, Recovery Procedures, determines what activities and tasks are to be taken, in what order, and by whom in order to affect the recovery. Section V, Appendices, contains all other information needed to carry out the plan.

B. Objectives

The objective of the Business Continuity Plan is to coordinate recovery of critical ZOS business functions in managing and supporting the business recovery in the event of a facilities disruption or disaster. This can include short or long-term disasters or other disruptions, such as fires, floods, earthquakes, explosions, terrorism, tornadoes, extended power interruptions, hazardous chemical spills, and other natural or man-made disasters.

A disaster is defined as any event that renders a ZOS business facility inoperable or unusable so that it interferes with the organization's ability to deliver essential business services. For ZOS, the critical service that must be maintained above all others is continuous Ground-to-Space Link (GSL) capability, which is the ability to command and control ZOS orbital assets. All BCP objectives flow from that requirement.

C. Scope and Applicability

This BCP applies to the ZOS Dallas ground station facility as the primary site. It covers the recovery of all IT infrastructure, orbital operations, personnel functions, financial operations, legal functions, and client communications necessary to sustain ZOS mission operations during and after a facility disruption. The Austin relay hub serves as the primary alternate site for GSL operations.

Section II: Business Continuity Strategy

A. Ground-to-Space Link Redundancy

ZOS operates orbital assets that require continuous ground command and control. The loss of the Dallas primary ground station without an active failover plan means loss of satellite command authority. This BCP is built specifically around maintaining GSL continuity through the Austin relay hub operated by Kevin D. Nguyen, Austin Relay Hub Operations Lead.

GSL failover procedures are classified as Priority One recovery activities. All other recovery activities are subordinate to maintaining orbital asset command capability. The Recovery Time Objective (RTO) for GSL failover activation is 4 hours from the declaration of a facility disruption.

B. Recovery Strategy Overview

ZOS will implement a hot/warm site recovery strategy. The Austin relay hub maintains near-real-time configuration synchronization with the Dallas primary ground station and can assume GSL operations within the 4-hour RTO. All other functions (administration, finance, HR, legal) will operate from remote work arrangements or the Austin site for up to 2 weeks. Recovery of the Dallas primary site is targeted within 3 to 7 days for IT infrastructure and up to 30 days for full facility restoration.

C. Business Impact Analysis Summary

The following ZOS functions have been assessed for criticality and recovery priority:

Function	Recovery Priority / RTO
Ground-to-Space Link (GSL) Command Operations	Priority 1 / 4 hours
Mission Telemetry Data Processing	Priority 1 / 4 hours

IT Infrastructure (core network, servers, SIEM)	Priority 2 / 24 hours
Client Communications and Data Relay Services	Priority 2 / 24 hours
Personnel Notification and HR Functions	Priority 3 / 48 hours
Financial Operations and Payroll	Priority 3 / 72 hours
Legal and Contract Management	Priority 4 / 5 business days
Administrative and Facilities Functions	Priority 5 / As feasible

D. Recovery Phases

Phase	Description
Phase I: Disaster Occurrence	Initial response to the disaster or disruptive event. Damage assessment, life safety, and initial notification of key personnel. GSL failover assessment begins immediately.
Phase II: Plan Activation	Notification and assembly of the BCP Recovery Team. Review and confirmation of the recovery strategy. Activation of the Austin relay hub if GSL continuity is threatened.
Phase III: Alternate Site Operations	Full operation from the Austin alternate site or remote work arrangements. GSL operations transferred to Austin. Client communications managed centrally.
Phase IV: Return to Normal Operations	Restoration of the Dallas primary site. Validation and testing of restored systems. Phased return of operations from Austin to Dallas. After-action review.

Section III: Recovery Teams

A. Purpose and Objective

This section identifies who will participate in the ZOS recovery process. The participants are organized into teams. Each team has a designated team leader and an alternate. Team members are assigned either to specific responsibilities or as members to carry out tasks as needed.

B. Recovery Team Assignments

Name / ZOS Title / BCP Role	Floor / Contact / Comments
Michael R. Torres Director of Mission Assurance and Continuity Incident Commander	Floor 3 m.torres@zos-secure.com Primary point of contact during all recovery operations; activates BCP; authorizes GSL failover
Daniel J. Kowalski Ground Systems IT Manager IT Recovery Lead and Alternate Team Leader	Floor 1 d.kowalski@zos-secure.com Leads IT infrastructure recovery; manages server and network restoration
Sandra L. Pham HR and Communications Director Public Information Officer and Alternate Team Leader	Floor 2 s.pham@zos-secure.com Handles internal and external communications; employee notifications; media liaison
Rachel M. Okafor Chief Financial Officer Finance and Admin Lead	Floor 4 r.okafor@zos-secure.com Manages financial continuity; emergency expenditure authorization; insurance claims
Gregory A. Whitfield General Counsel and Compliance Officer Legal and Contracts Lead	Floor 4 g.whitfield@zos-secure.com Oversees vendor and contract obligations during recovery; regulatory notifications
Lisa M. Cabrera Dallas Ground Station Operations Lead Business Function Leader (Dallas)	Floor 2 l.cabrera@zos-secure.com Coordinates Dallas site operations during recovery; liaison to facility management

Kevin D. Nguyen Austin Relay Hub Operations Lead Business Function Leader (Austin Failover)	Austin Site k.nguyen@zos-secure.com Activates and manages Austin relay hub GSL operations; assumes command authority during Dallas outage
James R. Holloway Satellite Network Administrator Network and Systems Recovery Lead	Floor 1 j.holloway@zos-secure.com Supports Kowalski with server and network recovery; ZOS network configuration restoration
Carlos E. Mendoza Orbital Systems Security Analyst Cybersecurity Recovery Monitor	Floor 1 c.mendoza@zos-secure.com Monitors systems for threats during recovery; verifies security posture before restoration
Patricia M. Solis Payroll and Accounting Finance Recovery Support	Floor 4 p.solis@zos-secure.com Assists Okafor with financial operations and payroll continuity
Thomas W. Brandt Facilities Manager Physical Site and Relocation Coordinator	Floor 1 t.brandt@zos-secure.com Coordinates physical site access and relocation logistics; vendor coordination for repairs
Angela C. Rivera HR Coordinator Employee Notifications	Floor 2 a.rivera@zos-secure.com Supports Pham with employee notifications and welfare tracking
Marcus L. Webb IT Support Technician End-User Recovery Support	Floor 1 m.webb@zos-secure.com End-user support during recovery; workstation setup at alternate site
Jennifer T. Walsh Operations Manager Day-to-Day Recovery Coordination	Floor 3 j.walsh@zos-secure.com Oversees day-to-day recovery task coordination across all teams
Donna K. Fields Executive Assistant 	Floor 3 d.fields@zos-secure.com Administrative support to Incident Commander Torres; scheduling and documentation

Administrative Support to Incident Commander	
Noe Valencia Principal Systems Security Engineer Security Recovery Coordinator	Floor 1 noe.valencia@zos-secure.com Security oversight of all recovery activities; coordinates with Huntress MDR during recovery; validates security posture before return to operations

C. Personnel Notification

Personnel notification follows the call tree structure defined in Appendix A. The Incident Commander (Michael R. Torres) initiates the call tree upon declaration of a disaster. If Torres is unavailable, Daniel J. Kowalski or Sandra L. Pham serve as alternate team leaders and initiate notification. Notification can also be made using reverse 911 or other mass notification systems. The Austin relay hub (Kevin D. Nguyen) is on a separate notification path to ensure GSL failover can proceed independently of Dallas site status.

Section IV: Recovery Procedures

Phase I: Disaster Occurrence

ACTIVITY: Initial Response and Damage Assessment

PERFORMED AT: Primary or Alternate Location

RESPONSIBILITY: Incident Commander / Senior Management on Site

1. The senior ZOS manager present at or first aware of the disaster activates emergency response and notifies the Incident Commander (Michael R. Torres) immediately.
2. If the disaster threatens immediate life safety, evacuate the facility per established ZOS emergency evacuation procedures before initiating BCP activities.
3. The Incident Commander assesses the nature and scope of the disaster and determines whether BCP activation is required.
4. Carlos E. Mendoza conducts an immediate cybersecurity assessment to determine if the disaster is accompanied by or caused by a cyber incident. If a cyber incident is suspected, activate the Incident Response Plan concurrently with BCP procedures.
5. Kevin D. Nguyen at the Austin relay hub is notified immediately of any event that may threaten Dallas GSL capability. Nguyen places Austin operations on standby for immediate GSL failover.
6. If GSL capability is threatened or lost, Nguyen activates Austin relay hub GSL operations without waiting for formal BCP declaration.

Phase II: Plan Activation

ACTIVITY: Notification and Assembly of Recovery Teams and Employees**PERFORMED AT: Alternate Site / Emergency Operations Center****RESPONSIBILITY: IT Management Team**

1. The Incident Commander calls each member of the management team, instructs them of what time frame to assemble at the Emergency Operations Center, and to bring their copies of the Plan. The EOC location is the Austin relay hub or an alternative location designated at the time based on accessibility.
2. Review the recovery strategy and action plan with the assembled team.
3. If necessary, adjust management team assignments based on which members are available.
4. The Management Team contacts critical employees and tells them to assemble at the alternate site. Non-critical employees are instructed to stay at home and work remotely until notified otherwise.
5. In the event of a disaster that affects telecommunications service regionally, the Management Team should instruct critical employees to proceed to the Austin alternate site even if they have not been contacted directly.
6. Daniel J. Kowalski assesses the status of all ZOS IT infrastructure and determines the recovery approach for servers, network devices, and communications systems.
7. Noe Valencia verifies the security status of all systems, confirms Huntress MDR is active and monitoring, and ensures no active cyber threat is complicating the recovery.

Phase III: Alternate Site Operations**ACTIVITY: Alternate Site Processing Procedures****PERFORMED AT: Austin Relay Hub / Remote Work Locations****RESPONSIBILITY: Alternate Site Operations Team**

1. Kevin D. Nguyen confirms full GSL operational capability at the Austin relay hub and notifies the Incident Commander that command and control of ZOS orbital assets has been successfully transferred.
2. Sandra L. Pham communicates with ZOS clients regarding the disaster, confirms continuity of satellite data relay services through the Austin hub, and manages all external communications.
3. Daniel J. Kowalski initiates IT infrastructure recovery from Veeam backups. Recovery sequence: (1) Domain controllers and Active Directory; (2) Core networking and firewall configuration; (3) Application servers; (4) File servers and data; (5) Communication systems.
4. Noe Valencia verifies the integrity of all backup data before restoration using Veeam backup integrity checks. No system is returned to production without security verification.
5. Thomas W. Brandt coordinates with facility management, insurance, and vendors for Dallas site damage assessment, cleanup, and repair timeline estimation.

- 6. Rachel M. Okafor activates emergency financial procedures, authorizes recovery expenditures, and coordinates with the cyber insurance provider if the disaster involved a covered cyber event.
- 7. Jennifer T. Walsh tracks all recovery tasks, maintains the recovery log, and provides status updates to the Incident Commander at defined intervals.

Phase IV: Return to Normal Operations

ACTIVITY: Restoration and Return to Primary Site
PERFORMED AT: Dallas Primary Facility (when restored)
RESPONSIBILITY: Full Recovery Team

- 1. Thomas W. Brandt confirms the Dallas primary facility is structurally safe, has restored utilities, and is ready for occupancy.
- 2. Daniel J. Kowalski conducts a full technical readiness assessment of all restored IT infrastructure before returning any systems to production. This includes network device configuration verification against baseline, server integrity validation, and Splunk SIEM operational confirmation.
- 3. Noe Valencia conducts a security validation assessment including: verification of all access credentials; confirmation that Huntress MDR is monitoring all endpoints; review of any security events that occurred during the recovery period; and sign-off on the security posture before operations transfer back to Dallas.
- 4. Kevin D. Nguyen coordinates the transfer of GSL operations back to the Dallas primary ground station after Kowalski and Valencia confirm full readiness. Transfer is executed during a low-traffic period for orbital assets.
- 5. A full after-action review is conducted within 5 business days of return to normal operations. The review covers: the cause and timeline of the disaster; effectiveness of the BCP activation; gaps identified in recovery procedures; and lessons to be incorporated into the next BCP revision.
- 6. The BCP is updated based on after-action review findings. All updates are distributed to recovery team members within 10 business days.

Section V: Appendices

Appendix A: ZOS Employee Telephone List

Employee / Title	Contact Information
* Michael R. Torres, Incident Commander	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx m.torres@zos-secure.com Team Leader
** Daniel J. Kowalski, Ground Systems IT Manager	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx d.kowalski@zos-secure.com Alt. Team Leader

** Sandra L. Pham, HR and Communications Director	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx s.pham@zos-secure.com Alt. Team Leader
Rachel M. Okafor, CFO	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx r.okafor@zos-secure.com
Gregory A. Whitfield, General Counsel	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx g.whitfield@zos-secure.com
Lisa M. Cabrera, Dallas Ground Station Operations Lead	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx l.cabrera@zos-secure.com Dallas Site
Kevin D. Nguyen, Austin Relay Hub Operations Lead	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx k.nguyen@zos-secure.com Austin Site
Marcus L. Webb, IT Support Technician	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx m.webb@zos-secure.com
Carlos E. Mendoza, Orbital Systems Security Analyst	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx c.mendoza@zos-secure.com
Angela C. Rivera, HR Coordinator	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx a.rivera@zos-secure.com
Thomas W. Brandt, Facilities Manager	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx t.brandt@zos-secure.com
Patricia M. Solis, Payroll and Accounting	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx p.solis@zos-secure.com
Jennifer T. Walsh, Operations Manager	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx j.walsh@zos-secure.com
Donna K. Fields, Executive Assistant	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx d.fields@zos-secure.com
James R. Holloway, Satellite Network Administrator	Office: 123-xxx-xxxx Mobile: 123-xxx-xxxx j.holloway@zos-secure.com

Noe Valencia, PSSE	Office: (214) 555-0147 Mobile: [Mobile] noe.valencia@zos-secure.com
Fire, Police, Emergency	911

* Indicates Team Leader | ** Indicates Alternate Team Leader

Appendix B: ZOS Alternate Site Information

Field	Details
Primary Alternate Site	Austin Relay Hub, ZOS Austin Operations, Austin, Texas
Austin Site Lead	Kevin D. Nguyen, Austin Relay Hub Operations Lead
Austin Site Contact	k.nguyen@zos-secure.com Mobile: 123-xxx-xxxx
Austin GSL Capability	Full Ground-to-Space Link command and control capability
Austin IT Infrastructure	Redundant network stack; Veeam backup restoration capability; full communications systems
RTO for GSL Failover	4 hours from Dallas facility disruption declaration
Secondary Alternate Site	Remote work arrangements for non-mission-critical functions

Appendix C: ZOS Recovery Priorities

Priority / Function	RTO / Description
Priority 1: GSL Command Operations	4 hours / Austin relay hub activation; transfer of orbital asset command authority
Priority 1: Mission Telemetry Processing	4 hours / Restore telemetry data processing capability at Austin
Priority 2: Core IT Infrastructure	24 hours / Domain controllers, core networking, Splunk SIEM, Huntress MDR verification
Priority 2: Client Data Relay Services	24 hours / Confirm defense and commercial client data relay continuity through Austin

Priority 3: Email and Communications	48 hours / Microsoft 365 is cloud-hosted and expected to be continuously available; internal communication systems restored
Priority 3: HR and Payroll Systems	72 hours / Access to ADP or equivalent payroll system; employee emergency support
Priority 4: Legal and Compliance Systems	5 business days / Access to Archer GRC, contract management, regulatory notification capability
Priority 5: Administrative Functions	As feasible / Non-critical administrative systems and facilities

Appendix D: Emergency Operations Center Locations

EOC Option	Location / Conditions
Primary EOC	Austin Relay Hub, ZOS Austin Operations Facility Preferred for any event affecting the Dallas primary facility
Secondary EOC	Virtual EOC using Microsoft Teams, Signal secure messaging, and external conference bridge For events where travel to Austin is not feasible
Tertiary EOC	Designated hotel conference room in the Dallas-Fort Worth metroplex For short-duration local events requiring in-person coordination

Appendix E: Vital Records Inventory

The following ZOS records are classified as vital records and must be recoverable within 24 hours of a disaster declaration. All vital records are backed up daily via Veeam Enterprise Plus to the ZOS offsite backup repository with copies at the Austin relay hub.

- Satellite command and telemetry configuration files
- Client contracts and service level agreements
- ZOS policy and procedure documents (stored in SharePoint Online via Microsoft 365)
- Personnel records and HR data
- Financial records, accounts, and payroll data
- Network device configurations (backed up nightly to Splunk and secure file storage)

- Cryptographic keys and certificates (stored in CyberArk PAM)
- Legal and regulatory compliance documentation (stored in Archer GRC)

Appendix F: ZOS BCP Maintenance Schedule

Activity	Frequency / Responsible Party
BCP Review and Update	Annual minimum; after any significant organizational change; after any BCP activation Noe Valencia (PSSE) with IRT
Recovery Team Contact Information Update	Quarterly Sandra L. Pham
Austin Relay Hub GSL Failover Test	Semi-annual Kevin D. Nguyen with Daniel J. Kowalski
Full BCP Tabletop Exercise	Annual Incident Commander with full Recovery Team
Veeam Backup Restoration Test	Quarterly Daniel J. Kowalski and James R. Holloway
After-Action Review (post-activation)	Within 5 business days of any BCP activation Incident Commander

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- Cisco Meraki. MR57 Access Point. <https://meraki.cisco.com>
- Huntress. Managed EDR Platform. <https://www.huntress.com>
- Splunk Inc. Splunk Enterprise SIEM. <https://www.splunk.com>
- Okta Inc. Workforce Identity Cloud. <https://www.okta.com>
- CyberArk. Privileged Access Management. <https://www.cyberark.com>
- Zscaler. Zero Trust Exchange. <https://www.zscaler.com>
- Qualys. Vulnerability Management, Detection and Response (VMDR). <https://www.qualys.com>
- KnowBe4. Security Awareness Training. <https://www.knowbe4.com>
- Veeam. Enterprise Plus Backup and Recovery. <https://www.veeam.com>
- Dell Technologies. PowerEdge R750 Server. <https://www.dell.com>